TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

GRADE 5

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · BAHÇEŞEHİR · İSTANBUL · İSTANBUL GALATA UNIVERSİTESİ

1. $\square \times (\square \times \square - \square) = 28$

By placing 4 of the digits 1, 2, 3, 4 and 5 in the boxes above, the given equality will be ensured.
If the digits in the boxes above different from each other what is the smallest digit which is no used?

A) 1 B) 2 C) 3 D) 4 E) 5

2.

- The number is an odd number
- The sum of the digits of the number is 7

In order for her friends to write the same number with Ayse definitely, at least how many number should they write?

A) 6 B) 8 C) 9 D) 10 E) 12

3. In a classroom 18 of the students are boys and $\frac{7}{13}$ of the students are girls.

According to the given informations above which of the followings is true for the number of girls and boys in this classroom?

A) The number of girls is 1 more than the number of boys
B) The number of girls is 2 more than the number of boys
C) The number of girls is 3 more than the number of boys
D) The number of boys is 3 more than the number of girls
E) The number of boys is 1 more than the number of girls

4. In the mathematics camp Ali and Ayşe is waiting in a line for lunch. Ali is the 8th from the beginning among all people in the line and is 3rd from the beginning among the boys. On the other hand Ayşe is the 14th from the beginning among all people in the line and is 6th from the beginning among the girls. According to the given information how many girls and boys are there in between Ali and Ayşe in the line?

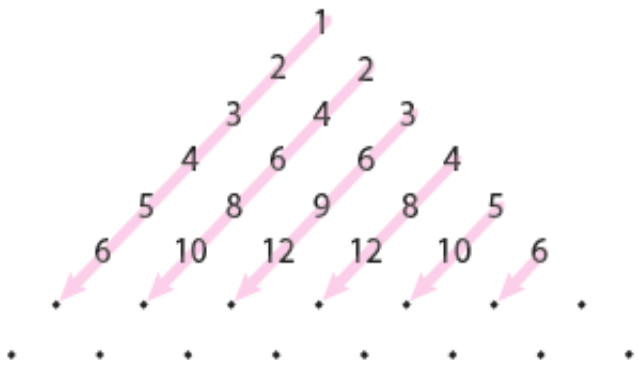
A) 1 girl, 4 boys B) 1 girl, 3 boys C) 2 girls, 3 boys
D) 2 girls, 4 boys E) No girl, 5 boys

5. Ceren choses two different numbers among the natural numbers from 1 to 12 each time and multiplies these numbers.

How many even numbers can she get by multiplying all possible pairs she choose?

A) 31 B) 32 C) 51 D) 52 E) 63

6. In the figure below, a triangle generated from numbers is given. In this triangle, the numbers are written according to a certain rule in the arrow direction.



According to this how many times the number 36 occurs in this triangle?

A) 18 B) 12 C) 9 D) 6 E) 4

7. In a math club out of 2 students 12 of them are boys. Each week two girls and one boy is accepted to the club as a member.
How many member will the club have in the day the numbers of girls and boys are equal to each other?

A) 32 B) 34 C) 36 D) 38 E) 40

8. Selma and Tanem play a game together with the following rules. First Selma writes a number to a paper then Tanem, in each of her consequtive move, erases that number and writes eather 2 more or 3 less than the last erased number.

If Selma writes 1923, how many of the numbers 1001, 1953, 2001 and 2024 can be obtainer by Tanem?

A) 1 B) 2 C) 3 D) 4 E) None of them

9. In a tournament between four football teams each team will play exactly once match with the other teams. In this tournament a team gets 3 point for a win, 1 point for a draw and no point for defeat. At the end of the tournament, teams' total points are 5, 4, 3 and 2.

According to above informations how many draws are there in this tournament?

A) 1 B) 2 C) 3 D) 4 E) 5

10. Deniz finds the average of 6 different positive integers as 7.
What is the maximum value of the greatest one of this 6 numbers?

A) 30 B) 29 C) 28 D) 27 E) 25

11. In a ticket sale point there are A, B an C counters. Tahir is the 5th person from the beginning in the line of the counter A, Ayşe is the 8th person from the beginning in the line of the counter B and Zekeriya is the 11th person from the beginning in the line of the counter C. At his time the number of the person behind of each of them in their line is twice the number of the person in front of them. Due to a problem on the counters B and C, first the persons in the line of B and then the persons in the line of C are added to the end of the line of the counter A.

In the final positon what is the Zekeriya's rank in the line of the counter A from the beginning?

A) 46 B) 47 C) 48 D) 51 E) 66

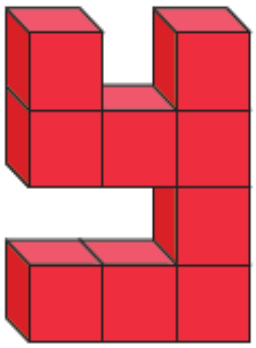
12. In the table below, the number of Arda and Kerem's goals in a season, the number of assists they had and the goal contribution numbers obtained by the sum of these numbers are given. (Assist: Pass resulting in a goal)

	Goal	Assist	Goal Contribution
Arda	18	7	25
Kerem	10	22	32

According to the information provided, maximum to how many goals did Arda and Kerem help together?

A) 22 B) 25 C) 32 D) 40 E) 57

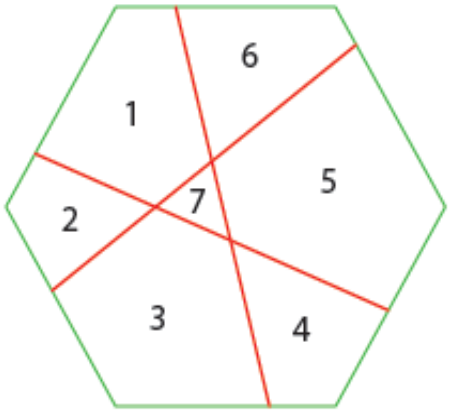
13. Yunus, by using the cubes whose edge length is 1 unit, wants to create a model of his name's first letter of his name shown in the figure below and then he wants to paint the model with red color.



If in order to paint one of the cube 9 liter paint is needed howmany liter of red paint is needed to paint the whole model?

A) 48 B) 54 C) 57 D) 63 E) 66

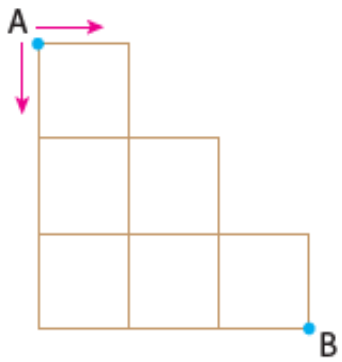
14. When the hexagon in the figure is cut along the 3 red lines 7 pieces can be obtained.



At most how many pieces could be obtained if this hexagon would cut by 4 red lines?

A) 9 B) 10 C) 11 D) 12 E) 13

15. An at the point A can only move right and down on the lines to get to the point B.



In how many diferent ways can the ant get to the point B from the point A?

A) 9 B) 10 C) 14 D) 15 E) 16

16. By writing each digit consecutively up to its value the following number is obtained.
122333444455…999

After than the number obtained is writen 1001 times side by side.

What is the digit at the middle of the final number?

A) 7 B) 6 C) 5 D) 3 E) 2

17. View of a cube from three different directions is given below.

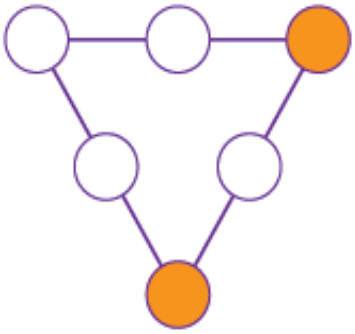
According to views given what is the shape on the surface of the cube opposite to the surface containing ● ?

A) ▲ B) ■ C) ▲ D) ● E) None of them, H

18. Given a counting number, if the sum of the positive divisors different from itself is greater than the given number than this number is defined as TMY number.
For instance for the positive divisors of 20 different than 20 are 1, 2, 4, 5 and 10. Since 1+2+4+5+10=22>20, 20 is said to be a TMY number.
According to the definition given above what is the su of the digits of the greatest 2-digit TMY number?

A) 17 B) 16 C) 15 D) 14 E) 13

19. Each of the numbers 1, 2, 3, 4, 5 and 6 is written once in the circles below.



In the figure the sum of the numbers in the circles on the sides of the triangle are equal to each other. According to this what is the smallest value of the product of the numbers in the colored circles?

A) 2 B) 6 C) 10 D) 12 E) 15

20. A bus starts its trip from İzmir and goes to İstanbul, Ankara and Antalya respectively and after arriving Antalya it comes back to İzmir directly. In the table below the number of passengers who gets on the bus in these cities and the number of passengers who gets off the bus in these cities are given.

Get on \ Get off	İzmir	İstanbul	Ankara	Antalya
İzmir		11	9	5
İstanbul	7		3	8
Ankara	2			13
Antalya	6			

Which of the following gives the number of the passengers on the bus between the cities?

A) İzmir $\xrightarrow{25}$ İstanbul $\xrightarrow{29}$ Ankara $\xrightarrow{42}$ Antalya $\xrightarrow{15}$ İzmir

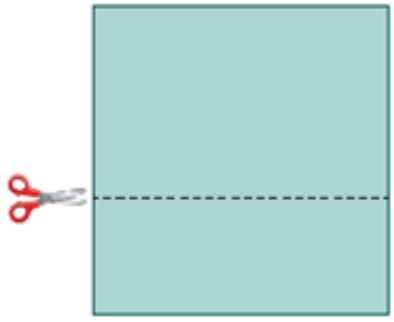
B) İzmir $\xrightarrow{25}$ İstanbul $\xrightarrow{32}$ Ankara $\xrightarrow{35}$ Antalya $\xrightarrow{15}$ İzmir

C) İzmir $\xrightarrow{25}$ İstanbul $\xrightarrow{29}$ Ankara $\xrightarrow{35}$ Antalya $\xrightarrow{15}$ İzmir

D) İzmir $\xrightarrow{25}$ İstanbul $\xrightarrow{35}$ Ankara $\xrightarrow{42}$ Antalya $\xrightarrow{15}$ İzmir

E) İzmir $\xrightarrow{25}$ İstanbul $\xrightarrow{32}$ Ankara $\xrightarrow{37}$ Antalya $\xrightarrow{15}$ İzmir

21.



A square is cut by a scissors into two rectangles as shown in the figure.
If the circumferences of the obtained rectangles are 26 cm and 34 cm what is the circumference of the square before it is cut?

A) 40 B) 44 C) 48 D) 52 E) 15

22. a, b and c are two digit natural numbers.
If the equality holds

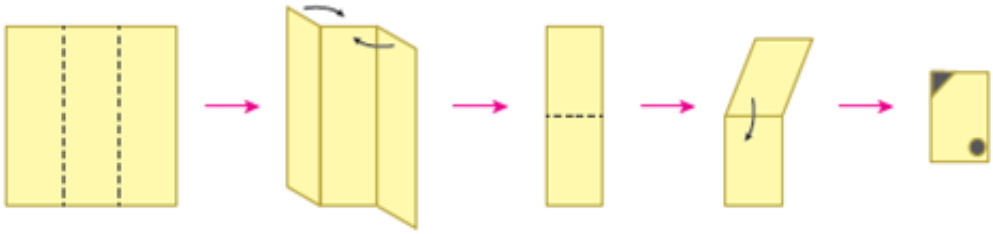
$$\frac{2024}{a-5}=\frac{2024}{b+3}=\frac{2024}{c-4}$$

Which of the following order is true for a, b and c?

A) $a<c<b$ B) $b<c<a$ C) $a<b<c$

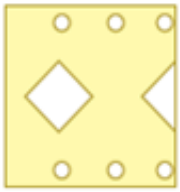
D) $b<a<c$ E) $c<a<b$

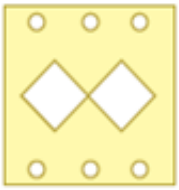
23. A square paper is marked by dashed lines into three equal rectangle.

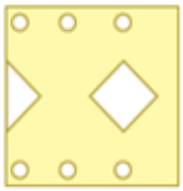


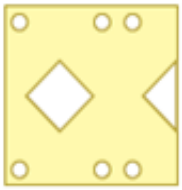
Then it is folded as shown in the figure below and the black part in the final fold is cut out.

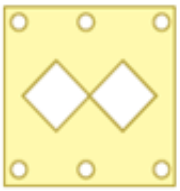
Which of the following is the opened form of the paper after the cut?

A) 

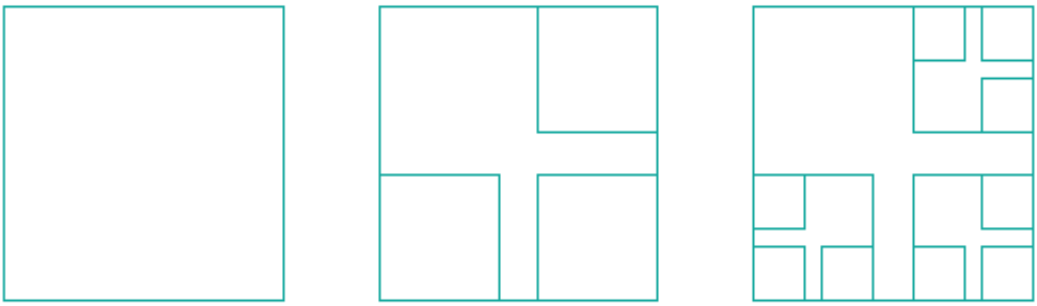
B) 

C) 

D) 

E) 

24. The first three steps of a pattern obtained by using a square are given above.



1.Step 2. Step 3. Step

How many square are there in the 5th step of this pattern?

A) 121 B) 112 C) 109 D) 98 E) 81

25. In the figure above the big rectangles are congruent and in each rectangle the shaded part represants a fraction.

What is the sum of the fractions represanted by the shaded parts of the rectangles?

A) $\frac{3}{4}$ B) $\frac{5}{4}$ C) $\frac{7}{9}$ D) $\frac{8}{9}$ E) 1